Given: Due:	Further Maths (Statistics) HW 09	Name:

Chapter 2: Poisson Distribution

You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.

91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)

Useful resources: *Useful resources*: Your notes from recent lessons & your text book

Questions - do these on lined paper

1) Accidents occur randomly at a road junction at a rate of 18 every year.
The random variable X represents the number of accidents at this road junction in the next 6 months.

(a) Write down the distribution of X.

(2)

(b) Find $P(X > 7)$.

(2)

(c) Show that the probability of at least one accident in a randomly selected month is 0.777 (correct to 3 decimal places).

(3)

2) A robot is programmed to build cars on a production line. The robot breaks down at random at a rate of once every 20 hours.
(a) Find the probability that it will work continuously for 5 hours without a breakdown.

(3)

Find the probability that, in an 8 hour period,
(b) the robot will break down at least once,

(2)

(c) there are exactly 2 breakdowns.

(1)

A new client wants to see the robot for themselves to be convinced of its reliability. What is the maximum duration the site manager can let the client see the robot in action and be at least 90% confident of a fault-free performance?

(3)

In a particular 8 hour period, the robot broke down twice.
(e) Write down the probability that the robot will break down in the following 8 hour period. Give a reason for your answer.

(2)

3) The number of heaters, H, bought during one day from *Warmup* supermarket can be modelled by a Poisson distribution with mean 0.7

(a) Calculate $P(H \geq 2)$

(1)

The number of heaters, G, bought during one day from *Pumraw* supermarket can be modelled by a Poisson distribution with mean 3, where G and H are independent.

(b) Show that the probability that a total of fewer than 4 heaters are bought from these two supermarkets in a day is 0.494 to 3 decimal places.

(2)

(c) Calculate the probability that a total of fewer than 4 heaters are bought from these two supermarkets on at least 5 out of 6 randomly chosen days.

(3)

4) An engineering company manufactures an electronic component. At the end of the manufacturing process, each component is checked to see if it is faulty. Faulty components are detected at a rate of 1.5 per hour.

(a) Suggest a suitable model for the number of faulty components detected per hour.

(1)

(b) Describe, in the context of this question, two assumptions you have made in part (a) for this model to be suitable.

(2)

(c) Find the probability of 2 faulty components being detected in a 1 hour period.

(2)

(d) Find the probability of at least one faulty component being detected in a 3 hour period.

(2)

5) The number of cars passing an observation point in a 10 minute interval is modelled by a Poisson distribution with mean 1.

(b) Find the probability that in a randomly chosen 60 minute period there will be

(i) exactly 4 cars passing the observation point,
(ii) at least 5 cars passing the observation point.

(4)

The number of other vehicles, other than cars, passing the observation point in a 60 minute interval is modelled by a Poisson distribution with mean 12.

(c) Find the probability that exactly 1 vehicle, of any type, passes the observation point in a 10 minute period.

(3)

END OF HOMEWORK
Total = 38 marks

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.